

 اونیورسیتی ملیسیا فہق السلطان عبد اللہ UNIVERSITI MALAYSIA PAHANG AL-SULTAN ABDULLAH	FACULTY OF COMPUTING		MARKS:
	COURSE: Web Engineering	CODE: BCS2243	
	TOPIC: Chapter 1-9	SEM: II 2024/2025	
	ASSESSMENT: Project	DURATION: 13 weeks	

General Instructions:

- Each group can have a maximum of 4 members.
- Please refer:
 - Table 1** for the project activities and datelines of submission.
 - Appendix 1** for the project description.
 - Appendix 2** for the rubric evaluation.
 - Appendix 3** for the cover page for each report submission.
- Any late submission or plagiarism will be penalized.**

Course Outcome:

The Course Outcome (CO) for this assessment as followed:

Course outcome	Outcome description
CO1	Design appropriate solution using fundamental web engineering concepts.
CO2	Construct a web-based application prototype using HTML, web server, database and scripting language based on web engineering methodology.
CO3	Demonstrate communication effectively in written and oral form through group discussion, meeting and presentation session.

The CO and marks distribution is summarised in Table 1:

Table 1. Assessment percentage according to CO

Assessment type		Course outcome percentage			
		CO1	CO2	CO3	Total (%)
Proposal		5	5	0	10
Progress	Progress 1	0	2	0	5
	Progress 2	0	3	0	
Project	Prototype	7	8	0	15
	Presentation	0	0	10	10
Grand total (%)					<u><u>40</u></u>

Question:

You are required to develop a web application prototype. The requirement of the application is attached in the **Appendix 1**. Table 2 summarised the project activities and submission.

Table 2. Project activities and submission dateline

Week	Day/Time	Assessment percentage	Milestones
7	25 April 2025	10%	Project proposal submission
9	Lab session C2: 12 May 2025* C1: 13 May 2025 C3: 15 May 2025 (Changes will be announced by the lecturer)	2%	Progress presentation 1: Development environment setup Database created Web page design Login module
11	Lab session C2: 26 May 2025 C1: 27 May 2025 C3: 29 May 2025 (Changes will be announced by the lecturer)	3%	Progress presentation 2: User session Manage data (insert, update, delete, view)
13	Lecture and Lab session 9 – 13 June 2025 (Exact date will be announced by the lecturer)	25%	Complete prototype from deployment server Final report submission

Students are required to prepare a proposal consists of the following items. The detailed evaluation is attached in Appendix 2.

1. The idea / principle of the solution
2. Review of TWO EXISTING SYSTEMS that similar to your idea
3. Project Management Framework
4. Project Requirements
5. Proposed Design (diagram required: use case diagram, use case scenarios hypertext structure model and access model)
6. Data Design (Entity Relationship Diagram (up to 3NF) and data dictionary)
7. Presentation model for each module (Graphical user interface design)
8. Testing Plan
9. Deployment Plan

Appendix 1

Project name: Fakulti Komputeran Petakom (MyPetakom)

The Faculty of Computing, Universiti Malaysia Pahang Al Sultan Abdullah (UMPSA) has established a student association named *Persatuan Teknologi Komputer* (Petakom). The association coordinates and manages various events and activities for the faculty students, which are eligible for co-curricular merits. However, many processes related to the association, such as membership registration, are still performed manually. Your software development team is responsible to design a web-based solution to manage the administrative tasks of Petakom, named MyPetakom. You are required to hold progress meetings with stakeholders every three weeks until the application is delivered. The basic requirements for the project are:

Module 1: Login, Manage Membership

1. *Login*

The login page with a web session is designed for different types of users: registered students, Event Advisor and Petakom Coordinator (administrator), the latter two of whom are also FK staffs. Users will see their registered names displayed on the page once they have successfully logged in.

2. *Manage membership*

All FK students are eligible to apply Petakom membership. Students must upload a copy of their student card for status verification, which will then be approved by the Petakom Coordinator.

3. *User profile management*

The module allows the system administrator and account owners to manage user profiles, including creating new users, updating existing profiles, deleting users and viewing profile information.

4. *Administrator dashboard*

The student responsible for this module is also assigned to develop the **Petakom Coordinator** dashboard. Student should collaborate with the student in Module 3 to identify different yet relevant information that facilitates calculations and generates insightful reports in graphical formats, such as graphs and charts.

Module 2: Manage event registration

1. *Event registration*

The Event Advisor is responsible for managing all events assigned to them, including creating, updating, deleting and viewing. In some cases, the registered events may be postponed or cancelled. MyPetakom will display this information in the application. The Event Advisor must specify the event's geolocation and upload the approval letter during the registration process.

2. *Event committee management*

The Event Advisor must also register the event's corresponding committees. Once the committees are registered, students assigned as the committee will be able to view the event details and position in their accounts. The recorded position is important for merit awards.

3. *Merit application for an event*

The Event Advisor must apply for student participation merit during the registration to ensure that the event qualifies for a merit award. The merit application will be approved by the Petakom Coordinator.

4. *Event QR code*

Once the event is completely registered, the application will generate a QR code for a specific event. Scanning the code will display a page containing the event information.

5. *User dashboard*

The student responsible for this module is also assigned to develop the MyPetakom **Event Advisor** dashboard. Student should identify relevant information that facilitates calculations and generates insightful reports in graphical formats, such as graphs and charts.

Module 3: Manage event attendance

1. *Manage event attendance*

The Event Advisor is responsible for creating an event attendance slot in MyPetakom based on the recorded event. Upon creation, MyPetakom will generate a QR code for the slot. Scanning this code will display a page containing the event attendance registration.

2. *Attendance registration*

All participants are required to scan the generated QR code for attendance registration. MyPetakom will display the geolocation and event description details. For security purposes, student who scan the QR code will be required to enter their student ID and password to verify their attendance (check in). MyPetakom will verify that the geolocation of the attendance matches the geolocation provided during event registration.

3. *Administrator dashboard*

The student responsible for this module is also assigned with developing the **Petakom Coordinator** dashboard. You should collaborate with the student in Module 1 to identify **different yet relevant** information that facilitates calculations and generates insightful reports in graphical formats, such as graphs and charts.

Module 4: Manage merit

1. *Manage missing merit*

Students can check the merits awarded for events and activities they participated through the application. The application allows student to claim missing merits, update and delete the claim application, as long as the claim status is not changed to "Submitted". Each claim application requires

student to upload an official letter for event participation. All claim applications will be approved by the Event Advisor. Students will be able to see the claimed merits in their merit list once approved.

2. *Merit awarded*

The application will calculate the cumulative merit for each semester based on a student's involvement in Petakom events or activities, either as a committee member or participant. The merit score for event committee is provided in Table 1. The cumulative merits will be displayed in each student's account. Additionally, a QR code will be generated for each student. Scanning the code will display a page with their basic information and total merits obtained for the academic year.

3. *MyPetakom student dashboard*

The student assigned to this module is also responsible for developing the MyPetakom **student** dashboard. The student should identify relevant information that allows for calculations and generates insightful reports in graphical formats, such as graphs and charts.

Table A. The merit scores for event committees

Event level	Main committee	Committee	Participant
International	100	70	50
National	80	50	40
State	60	40	30
District	40	30	15
UMPSA	30	20	5

General guidelines for project development and presentation:

1. Each group member is responsible for a specific module and required to:
 - a. propose a detailed process
 - b. develop the *CRUD* functionality (create, read, update and delete)
 - c. develop appropriate reporting for the module. The report must include a single table and a join table using Structured Query Language (SQL) and they must be distinct from each other. Repetition of calculations and report features will result in a deduction of marks.
2. All modules must be integrated with one another and presented as a complete web application.
3. The final presentation (complete prototype) is required to be hosted on Indah server or any free web hosting platform.
4. **Absence from the final presentation will result in a zero mark.**
5. You are recommended for a backup strategy by uploading your documents to online storage options (such as OneDrive, Google Drive, email or free web hosting).
6. A marks penalty will be applied for late submissions, while a score of zero will be given for failure to submit.

Appendix 2: Rubric Evaluation

2.1 Peer Review

Peer review will be evaluated using a google form make available to you in KALAM upon your submission date. The evaluation items for peer review are listed in Table 1. Each of the group member must fill in peer review form once it is available in KALAM.

Table 1. BCS2433 WEB ENGINEERING PROJECT: PEER EVALUATION FORM SEM II 2024/2025

Course Learning Outcomes:	
CLO3: Work effectively in group and promote leadership's	PLO4: Communication Skills

PEER EVALUATION RUBRICS						
ITEM		SCORE				
		0	1	2	3	4
A.	CONTRIBUTION	Very Low	Low	Moderate	High	Very High
	Contribute to the effort in writing as well as in team	Your partner had contributed 0%-19% of the report writing and and team effort (eg: absent in every meeting @ attend meeting but quiet) - sleeping partner	Your partner had contributed 20%-49% of the report writing and and team effort (eg: partially attend and team has to do for him/her part)	Your partner had contributed 50%-69% of the report writing and and team effort (eg: attend meeting but team has to cover his/her part)	Your partner had contributed 70%-89% of the report writing and and team effort (eg: attend all meetings but his/her report was submitted late)	Your partner had contributed 90%-100% of the report writing and and team effort (eg: full attendance in every meeting and complete his/her part in report)
B.	PARTICIPATION	Very Passive	Passive	Normal	Active	Very Active
	Participate actively in leading / facilitating discussion / team meetings / developing ideas and planning project (every meeting must provide logbook and snap group photo - as an evidence)	Your partner did not participate at all (eg: empty record in logbook)	Your partner participated above 20% of all activities in group	Your partner participated in between passive and active. Sometimes passive and sometimes active. (50% attended)	Your partner participated actively in most activities	Your partner participated actively in all activities in your group
C.	COLLABORATION	Entry Level	Adoption Level	Adaptation Level	Infusion Level	Transformation

	Worked cooperatively with other group members regards races, genders and seniority	Your partner was unable to collaborate in your group although he/she exists.	Your partner had taken action to collaborate but not in depth	Your partner had certain blending with all team members between the given expectation	Your partner had ability to mixture and collaborate with all team members	Your partner had transformed not only him / herself but also other member in class
D.	ATTITUDE	Very Negative	Negative	Neutral	Positive	Very Positive

	Displayed positive approach and made constructive comments in working toward goal	Your partner showed extreme negative feedback (eg: not interested to complete his/her tasks and create hatred and sabotage your	Your partner showed negative feedback (eg: not interested to complete his/her tasks only)	Your partner had neutral feedback which did not showed any extreme positive or negative actions	Your partner has shown positive feedback in his/her actions in team	Your partner had shown extremely positive not even in his/her group but also in overall member in the class
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PEER EVALUATION RUBRICS

ITEM		SCORE				
		0	1	2	3	4
E	COMMUNICATION	Never	Rarely	Sometimes	Frequent	Always
	Expressed thoughts clearly and show efforts in communicating well within group	Your partner was blurred in all discussions	Your partner was unable to give clear explanations and sometime can talk but the fact is not true	Your partner was able to give some parts of clear explanations	Your partner was able to deliver his / Her understanding within knowledge in teaching learning	Your partner was consistently deliver his communication with fluent and good knowledge (eg: the way he / she talk is truth and polite)
F	RESPONSIVENESS	Unresponsive	Responsive to	Responsive to	Awake	Alert
	Reacted sensitively to verbal and nonverbal cues of other team members within group	Your partner was unable to respond for any request in his / her tasks	Your partner was unable to respond verbal	Your partner was unable to respond nonverbal	Your partner was able to respond to verbal and nonverbal with good respond	Your partner was able to respond to verbal and nonverbal with advanced respond

USING THE ABOVE RUBRICS, EVALUATE YOUR TEAM MEMBER (By giving appropriate scores)

GROUP NAME:		EVALUATION CRITERION (BASED ON ABOVE RUBRIC RANK 0 TO 5)						
YOUR NAME (STD ID):								
YOUR TEAM MEMBER:								
No	Name (Std ID)	A	B	C	D	E	F	TOTAL
1								
2								
3								
4								

Appendix 3: Sample Front Page Cover



FACULTY OF COMPUTING

BCS2243 WEB ENGINEERING SEMESTER II 2024/2025

TITLE : *(Project Name)*
SECTION : **01 / 02 / 03** *(Remove unnecessary info)*
LECTURER : *(Your lecturer's name)*

Student Detail: *(Adjust all photos to fit in one page)*

Name	Student ID	Student Photo

-END-